Mengke Zhao

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Introduction

I am a postdoctoral at Nanjing University, working with Prof. Keping Qiu. I received my Ph.D. degree from Xinjiang Astronomical Observatory (XAO), Chinese Academy of Sciences (CAS) in 2024. My research interests center on magnetic fields, turbulence, and gravity in the interstellar medium's complex astrophysical processes and star formation. My research objects are interstellar medium in molecular clouds, massive star-forming regions, and nearby galaxies. I do my postdoctoral research at the Astronomy Department of Nanjing University on three topics:

- Application of Constrained diffusion algorithm, mapping density and scale, reconstructing three-dimensional density structure
- Complex astrophysical processes between magnetic field, gravity, and turbulence at multi-scales
- Rebuilding three-dimensional magnetic field structure

Education

Xinjiang Astronomy Observatory, Chinese Academy of Sciences *Urumqi, China*

Ph.D in Xinjiang Astronomy Observatory,CAS, Astrophysics 2020-2024

Xinjiang Astronomy Observatory, Chinese Academy of Sciences *Urumqi, China*

MS in Xinjiang Astronomy Observatory,CAS,Astrophysics 2018-2020

Journal Articles

Equation vs. AI: Predict Density and Measure Width of molecular clouds by Multiscale Decomposition, **Zhao Mengke**, Li Guang-Xing, Xu Duo, Qiu Keping [Link]

Slope of Magnetic Field–Density Relation as an Indicator of Magnetic Dominance, **Zhao Mengke**, Li Guang-Xing, Qiu Keping 2024, ApJ [Link]

Super-Jeans fragmentation in massive star-forming regions revealed by triangulation analysis, Li Guang-Xing, **Zhao Mengke**, Lu Xing, 2025 MNras [Link].

Magnetic, Kinetic, and Transition Regime: Spatially-Segregated Structure of Compressive MHD Turbulence , Li Guang-Xing, **Zhao Mengke**, 2025 MNras [Link].

Mapping Lorentz Force in Orion A, **Zhao Mengke**, Li Guang-Xing, Qiu Ke-Ping [Link].

The impact of expanding HII regions on filament G37: Curved magnetic field and multiple direction material flows, **Zhao Mengke**, Tang Xindi, Qiu Keping et al. 2025, A&A, [Link].

Magnetic Field in Giant Filaments Probed by Velocity Gradient Technique: Parallel Magnetic Field and the Existence of Magnetization Gaps, **Zhao Mengke**, Li Guang-Xing, et al, 2024, ApJ 961 124, [Link].

Magnetic Field of Molecular Gas Measured with The Velocity Gradient Technique II. kpc-pc bubble in NGC 628, **Zhao Mengke**, Zhou Jian-Jun, Baan Willem, Hu Yue, Lazarian Alex, Tang Xin-Di. et al, 2024, ApJ 961 124, [Link].

Magnetic Field of Molecular Gas Measured with the Velocity Gradient Technique I. Orion A, **Zhao Mengke**, Zhou Jian-Jun, Hu Yue, Lazarian Alex, Baan Wiilem, Tang Xin-Di. et al, 2022, ApJ, 934, 45 [Link].

Activity

As PI, awarded a Young Scientists Fund project (Category C) by the National Natural Science Foundation of China (NSFC)

STar-forming Region Observations (BISTRO) survey second general program (PI: Kate, Pattle).

STar-forming Region Observations (BISTRO) survey 3rd HMPO tiger team (PI: Keping, Qiu).

As PI, my JCMT cycle 21B proposal, observed by Scuba-2/Pol-2 at Band 1, was accepted.

As co-PI together with my supervisor Prof. JianJun Zhou, we have an accepted JCMT proposal for cycle 22A.

Technical Skills

Programming	Python and Julia
Data Analysis	Python and Gildas for data reduction of spectral lines
Computations	Multi-threaded computing in e.g. Julia
Computational astrophysical diagnostics	Velocity Gradient Technique for study of MHD turbulence and magnetic field in the ISM; mathematical tools,such as correlation functions for study of turbulent motions and gravitational fields.

Hobbies

Mountain Biking, Piano playing.